ROBERT AFIAKWA OWUSU

Biomedical Engineer | Computational Bioengineering | AI in Healthcare

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PROFESSIONAL SUMMARY

Biomedical engineer with research experience in molecular docking, AI-based prediction, and computational analysis of biological systems. Interested in molecular simulation, biomolecular interactions, and computational approaches to therapeutic discovery.

EDUCATION

Kwame Nkrumah University of Science and Technology, Ghana **Jan 2022 – Nov 2025**

B.Sc. Biomedical Engineering, First Class Honours

- GPA: 3.98/4.00; Valedictorian, Department of Computer Engineering.

RESEARCH INTERESTS

Computational Drug Discovery; Molecular Modeling and Simulation; AI/ML for Drug Safety; Protein Structure Modeling; Biomedical AI.

RESEARCH EXPERIENCE

Research Assistant | Global Health AI & Compute Lab (GHAiC-K), KCCR **Oct 2025 – Present**

Supervisors: Prof. Edmund Ekuadzi, Prof. John H. Amuasi, and Dr. Prince Ebenezer Adjei

- **Development of an AI-Enabled Systems Pharmacology Database for West African Medicinal Plants:** Leading the development of the West African Traditional Medicine System Pharmacology (WATM-SP) platform, integrating phytochemical, molecular, protein-target, disease, ADMET, ethnomedicinal, and therapeutic indication data into a unified computational framework. Collaborating with herbal medicine and pharmacy researchers on phytochemical screening, compound validation, and biological interpretation (*manuscript in preparation*).
- **Curation and Standardization of an ACE Bioactivity Dataset for Machine-Learning–Based Antihypertensive Discovery:** Building a curated ACE bioactivity dataset by extracting, harmonizing, and standardizing records from ChEMBL and PubChem to support machine-learning model development, ligand-based analysis, and computational lead prioritization.

Research Trainee | Spark Academy Imaging Research Program **Jan 2026 – Jun 2026**

- Co-authored a study on the robustness and reliability of deep-learning models for breast ultrasound lesion segmentation under progressive speckle corruption, benchmarking U-Net, Attention U-Net, TransUNet, and SegFormer-B0; manuscript submitted to the **MICCAI 2026 MIRASOL Workshop**.

Undergraduate Thesis | Low-Cost Portable Wireless Digital Stethoscope **Jan 2025 – Aug 2025**

Supervisor: Dr. Prince Ebenezer Adjei

- Developed a portable digital stethoscope integrating biomedical signal acquisition and wireless transmission for remote auscultation in resource-limited clinical environments.
- Collaborated with clinicians to evaluate device usability and refine the prototype for cardiovascular monitoring and telemedicine.

Research Assistant | Machine Learning Model Applications in Medicine **Sep 2024 – Oct 2024**

Supervisor: Dr. Daniel Akwei Addo

- Contributed to review work on machine-learning model applications in cardiovascular medicine and artificial intelligence models for predicting outcomes of lung pathologies.
- Conducted literature searches, synthesized evidence across biomedical studies, and assessed model performance, interpretability, generalizability, and dataset limitations.

SELECTED COMPUTATIONAL RESEARCH PROJECTS

Validation-Gated Structure-Based Screening of *Plasmodium falciparum* DHFR **Jun 2026 – Ju 2026**

- Developed and validated a Meeko/AutoDock Vina docking workflow, reproducing the crystallographic WR99210 pose with a heavy-atom RMSD of **1.51 Å** before applying the protocol to a nine-compound screen.

AI-Based Protein Structure Prediction and Structural Validation **Jul 2026-Aug 2026**

- Predicted human DHFR using AlphaFold3 and validated the model against experimental 1U72 using PyMOL, obtaining an RMSD of **0.375 Å** across 186 aligned residues.

Machine-Learning Cardiotoxicity Screening of West African Medicinal-Plant Compounds Dec 2025 – Feb 2026

- Developed and evaluated machine-learning models for hERG, Nav1.5, and Cav1.2 cardiotoxicity prediction and applied them to screen West African medicinal-plant compounds for potential cardiac ion-channel liabilities.

PUBLICATIONS AND MANUSCRIPTS

- "Beyond Dice: Evaluating Reliability and Robustness in Breast Ultrasound Lesion Segmentation Under Progressive Speckle Corruption." Submitted to the **MICCAI 2026 MIRASOL Workshop**.
- "West African Traditional Medicine System Pharmacology Platform." *Manuscript in preparation.*

TECHNICAL SKILLS

Research and Data Analysis: Scientific writing, study design, data curation, statistical analysis, data visualization, and reproducible computational workflows.

Computational Methods: Molecular docking, virtual screening, protein–ligand interaction analysis, protein structure evaluation, machine-learning classification, and introductory molecular dynamics simulation.

Programming and Molecular Modeling Tools: Python, pandas, NumPy, scikit-learn, Linux/WSL, command-line scripting, AutoDock Vina, Meeko, PyMOL, UCSF Chimera, DoGSiteScorer, AlphaFold3, and GROMACS.

TEACHING AND PROFESSIONAL EXPERIENCE

Teaching Assistant | Department of Computer Engineering, KNUST — **Oct 2025 – Present**

- Selected as Teaching Assistant based on outstanding academic performance to support Biosignals Analysis, Digital Signal Processing, and Biomedical Optics through tutorials, laboratory instruction, instructional material development, and undergraduate capstone mentoring.

Biomedical Engineering Intern | Komfo Anokye Teaching Hospital — **Sep 2023 – Nov 2023**

- Performed medical-equipment inspections, preventive maintenance, troubleshooting, inventory documentation, and clinical-user support.

Biomedical Engineering Intern | Swedru Municipal Hospital — **Oct 2022 – Dec 2022**

- Assisted with medical-equipment installation, routine inspection, maintenance, troubleshooting, and inventory management.

CONFERENCES & PRESENTATIONS

- ***A Low-Cost Portable Wireless Digital Stethoscope for Clinical Use in Ghana***
  Poster Presentation, *Noguchi Annual Research Meeting*, Accra, Ghana — Nov 2025
- ***A Low-Cost Portable Wireless Digital Stethoscope for Clinical Use in Ghana***
  Poster Presentation, *Technology Week, KNUST*, Kumasi, Ghana — Aug 2025
- **Frontiers in Optics + Laser Science Conference** (Optica)
  Conference Participant, Denver, Colorado, USA — Oct 2025

HONORS AND AWARDS

- **Valedictorian, Department of Computer Engineering, KNUST** — Nov 2025
- **Best Biomedical Engineering Student, College of Engineering Provost Awards** — Jul 2023 – 2026
- **Optica Amplify Immersion Program Scholar, Denver, Colorado, USA** — Oct 2025
- **Winner, Ghana National Gas Challenge** — Nov 2024
- **Ghana Education Trust Fund Scholar** — Apr 2022
- **Best Graduating Student, Sweru Senior High School** — Nov 2020
- **Winner, Second Ghana Science Olympiad** — Dec 2019

LEADERSHIP AND PROFESSIONAL MEMBERSHIPS

Academic Head, Ghana Engineering Students Association; Clinic Project Team Lead, Engineers Without Borders–KNUST; Former Vice President, Biomedical Engineering Students Society. Member of Optica, SPIE, NSBE, and Black in AI.